V3 empirical study · Instruction

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of seven models tracks realized stock price S_t most closely, and does the ranking change across companies and the four volatility regimes?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t. That row is bold and shaded green; Mark ranks 1–7.

Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY, AAPL, MSFT, AMZN. Each name is calibrated and simulated separately.
- Regimes: Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t0 are used only after t0.
- P-measure: keep the lookback drift μ. Do not replace μ → r (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

Seven models · four underlyings · four volatility regimes · 1.5-year fixed calibration

Path construction

- One rolling path cloud per ticker × regime × model, n_paths = 10000, seed 42.
- Clock: daily trading days from the first bar of the regime through the regime end. $\Delta t = 1/252$.
- Start at the observed open S_0. Hold the single t0 calibration for every day of the regime.
- At each day, summarize the cloud by p10, p25, p50, p75, p90. The reported path is p50, not the Monte Carlo mean.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. Ranking uses this only.
- $\text{MAE} = \text{mean}\,|p50 - S_t|$. Bias \$ = $\text{mean}(p50 - S_t)$. Bias% = $100 \times \text{mean}((p50 - S_t)/S_t)$.
- ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75 − p25). ICP 10–90 is recorded but not a table column.
- ICP is higher-better and is not used for the green row.

Page 3 · Table block · SPY · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
GBM	34.41	25.9924	+25.4098	+29.70	11.1%	19.6746	3
Modified GBM	35.81	27.1703	+26.6664	+31.11	9.9%	18.2026	5
GARCH	37.95	28.9693	+28.4668	+33.17	11.9%	22.0506	6
Heston	34.63	26.1716	+25.6502	+29.95	13.1%	22.1563	4
Merton	**30.85**	**22.9129**	**+22.2314**	**+26.10**	**14.7%**	**19.0387**	**1**
GARCH–Merton	39.77	30.4684	+30.0004	+34.91	11.9%	22.8520	7
Heston–Merton	33.94	25.5642	+25.0144	+29.24	14.3%	23.0305	2

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
GBM	5.00	8.1840	+8.1743	+4.20	81.4%	21.1529	3
Modified GBM	5.34	8.8286	+8.8206	+4.53	76.7%	21.5379	4
GARCH	8.35	14.0992	+14.0952	+7.19	30.4%	21.8668	7
Heston	4.63	7.5225	+7.5141	+3.86	90.9%	28.1731	2
Merton	6.14	10.2893	+10.2846	+5.27	61.7%	21.4296	5
GARCH–Merton	7.62	12.8551	+12.8530	+6.57	42.7%	22.7237	6
Heston–Merton	**4.41**	**7.1845**	**+7.1767**	**+3.69**	**92.5%**	**29.5426**	**1**

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
GBM	11.83	31.0558	+31.0558	+11.16	1.2%	31.1821	2
Modified GBM	**7.97**	**17.9884**	**+17.9884**	**+6.60**	**46.6%**	**29.2245**	**1**
GARCH	20.33	54.1256	+54.1256	+19.18	1.2%	31.5498	7
Heston	12.95	34.2496	+34.2496	+12.29	14.3%	41.8944	4
Merton	18.97	50.7657	+50.7657	+18.03	1.2%	32.5920	6
GARCH–Merton	16.51	44.1759	+44.1759	+15.73	1.2%	32.8409	5
Heston–Merton	12.75	33.6382	+33.6382	+12.08	25.1%	46.4233	3

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25–75	ABW	Mark
GBM	8.54	20.6229	-9.5160	-2.50	49.2%	41.3009	3
Modified GBM	9.21	23.6787	-15.6556	-4.50	41.7%	41.5597	5
GARCH	14.71	29.8725	+26.1417	+9.14	49.2%	48.3802	7
Heston	**8.45**	**20.1395**	**-8.7470**	**-2.25**	**61.9%**	**50.6964**	**1**
Merton	8.58	18.2211	-1.2970	+0.18	59.1%	41.8094	4
GARCH–Merton	12.31	24.9973	+18.3131	+6.57	65.5%	49.7937	6
Heston–Merton	8.50	20.5530	-9.7264	-2.57	65.5%	55.6121	2

RMSE% is 100×√mean(((p50−S_t)/S_t)²). MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 4 · Table block · AAPL · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	73.54	2.1069	+2.0402	+64.72	4.8%	2.1188	3
Modified GBM	**71.03**	**2.0176**	**+1.9533**	**+62.24**	**5.2%**	**1.9370**	**1**
GARCH	104.17	3.1299	+3.0758	+94.17	5.2%	2.5921	6
Heston	74.86	2.1495	+2.0862	+66.07	4.4%	2.0793	4
Merton	79.75	2.3207	+2.2578	+70.92	4.4%	2.1865	5
GARCH–Merton	110.41	3.3318	+3.2803	+99.97	5.2%	2.7863	7
Heston–Merton	72.88	2.0787	+2.0148	+64.05	7.1%	2.2490	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	21.23	3.9335	-3.6702	-15.85	32.8%	4.3434	6
Modified GBM	27.69	5.1179	-4.9949	-21.93	27.3%	3.7517	7
GARCH	13.48	2.5723	-2.0165	-8.28	56.1%	4.6278	2
Heston	20.65	3.8501	-3.5020	-15.02	31.2%	4.2697	4
Merton	14.31	2.7277	-2.1412	-8.79	52.6%	4.7267	3
GARCH–Merton	**13.31**	**2.5465**	**-1.9428**	**-7.91**	**72.7%**	**5.4147**	**1**
Heston–Merton	21.14	3.9346	-3.5807	-15.34	33.2%	4.8528	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	40.68	17.0222	+17.0107	+37.95	1.6%	12.1123	4
Modified GBM	43.92	18.5121	+18.5009	+41.11	1.6%	12.5894	6
GARCH	47.64	20.1873	+20.1762	+44.65	1.6%	12.7520	7
Heston	39.61	16.5269	+16.5154	+36.90	1.6%	12.8894	3
Merton	**34.13**	**13.9045**	**+13.8915**	**+31.33**	**2.0%**	**11.2247**	**1**
GARCH–Merton	43.62	18.3666	+18.3554	+40.81	1.6%	13.4838	5
Heston–Merton	38.67	16.0817	+16.0691	+35.95	2.0%	14.5604	2

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	30.37	22.4802	-22.4802	-27.34	4.0%	13.7604	6
Modified GBM	32.61	24.1778	-24.1778	-29.46	1.2%	13.6055	7
GARCH	**14.15**	**9.8287**	**-8.5651**	**-9.99**	**50.8%**	**20.4279**	**1**
Heston	29.84	22.0482	-22.0482	-26.75	5.2%	14.2112	4
Merton	29.38	21.7077	-21.7077	-26.35	5.6%	13.7630	3
GARCH–Merton	14.45	10.0988	-8.9957	-10.46	54.0%	22.3912	2
Heston–Merton	30.34	22.4176	-22.4176	-27.20	7.1%	16.0911	5

RMSE% is $100\times\sqrt{\text{mean}(((p50-S_t)/S_t)^2)}$. MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 5 · Table block · MSFT · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	19.26	2.2887	+1.4334	+11.22	42.1%	4.1984	4
Modified GBM	22.72	2.5827	+2.0905	+15.54	34.5%	4.4448	7
GARCH	**16.91**	**2.1048**	**+0.8897**	**+7.63**	**54.8%**	**4.6635**	**1**
Heston	18.90	2.2544	+1.3660	+10.76	43.7%	4.4086	3
Merton	22.57	2.5699	+2.0563	+15.32	34.9%	4.3125	6
GARCH–Merton	21.83	2.5049	+1.9408	+14.55	44.8%	5.3805	5
Heston–Merton	18.62	2.2386	+1.2935	+10.29	48.4%	4.7345	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	9.16	2.9735	-2.7770	-7.11	74.3%	7.1367	5
Modified GBM	13.58	4.4580	-4.2900	-11.07	41.1%	6.7242	7
GARCH	4.32	1.3197	-0.9766	-2.38	94.9%	7.7229	2
Heston	8.69	2.7921	-2.5511	-6.48	79.4%	7.3569	4
Merton	6.12	1.9212	-1.6448	-4.11	94.1%	7.2214	3
GARCH–Merton	**2.86**	**0.8682**	**-0.0403**	**+0.09**	**95.3%**	**9.2508**	**1**
Heston–Merton	9.44	3.0321	-2.7972	-7.12	89.3%	8.5397	6

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	16.53	17.7039	+17.7039	+15.66	15.1%	24.6968	3
Modified GBM	18.89	20.5816	+20.5816	+18.03	1.2%	23.5056	5
GARCH	25.74	28.5092	+28.5092	+24.54	1.2%	25.8335	7
Heston	15.62	16.5393	+16.5393	+14.70	27.9%	26.1265	2
Merton	18.64	20.2820	+20.2820	+17.80	2.4%	24.6924	4
GARCH–Merton	23.41	25.8543	+25.8543	+22.37	1.2%	28.2292	6
Heston–Merton	**15.57**	**16.4389**	**+16.4389**	**+14.63**	**43.8%**	**30.0522**	**1**

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	12.14	17.8924	-16.6883	-9.02	64.3%	34.7272	4
Modified GBM	9.44	13.6285	-11.5096	-6.06	78.2%	33.4464	3
GARCH	**6.21**	**6.3104**	**+0.5708**	**+0.79**	**91.7%**	**39.4932**	**1**
Heston	12.33	18.1804	-17.0140	-9.20	66.7%	35.7022	5
Merton	14.58	21.6196	-20.9360	-11.45	47.6%	33.2491	7
GARCH–Merton	6.98	9.1410	-5.4056	-2.57	91.7%	40.2136	2
Heston–Merton	12.77	18.8424	-17.7774	-9.64	69.0%	38.8735	6

RMSE% is 100×√mean(((p50−S_t)/S_t)²). MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 6 · Table block · AMZN · seven models × four regimes

P-measure daily cloud, reported path = p50. Bold green = lowest RMSE% vs realized S_t. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	46.90	1.1152	+1.0618	+36.69	49.6%	2.1141	4
Modified GBM	114.01	3.4021	+3.3747	+102.32	10.3%	3.1462	7
GARCH	58.67	1.5579	+1.5130	+49.66	17.1%	2.3638	6
Heston	44.76	1.0376	+0.9768	+34.18	49.6%	2.0395	3
Merton	**29.61**	**0.6553**	**+0.1297**	**+9.51**	**61.1%**	**1.6009**	**1**
GARCH–Merton	56.44	1.4796	+1.4290	+47.23	37.7%	2.7035	5
Heston–Merton	40.52	0.8817	+0.8092	+29.21	57.5%	2.3991	2

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 253 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	49.05	6.9955	+6.9931	+43.67	6.7%	5.8701	4
Modified GBM	55.85	7.9140	+7.9130	+49.42	6.7%	6.1535	5
GARCH	59.80	8.4353	+8.4341	+52.68	6.3%	6.3861	6
Heston	47.41	6.7601	+6.7570	+42.21	5.9%	5.7008	3
Merton	**39.35**	**5.6550**	**+5.6498**	**+35.30**	**7.1%**	**5.4556**	**1**
GARCH–Merton	60.00	8.4580	+8.4567	+52.83	6.7%	7.2906	7
Heston–Merton	46.00	6.5541	+6.5494	+40.92	5.9%	6.6082	2

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 251 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	51.83	42.8456	+42.8456	+48.55	0.4%	29.8080	4
Modified GBM	62.88	51.4866	+51.4866	+58.13	0.4%	34.2508	5
GARCH	72.29	58.6624	+58.6624	+66.05	0.4%	34.3867	7
Heston	50.24	41.5359	+41.5359	+47.10	0.4%	30.4324	2
Merton	50.76	41.9602	+41.9602	+47.56	0.4%	28.3860	3
GARCH–Merton	68.07	55.4059	+55.4059	+62.44	0.4%	37.8848	6
Heston–Merton	**48.85**	**40.4369**	**+40.4369**	**+45.87**	**0.4%**	**34.6482**	**1**

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 252 days

Model	RMSE%	MAE	Bias $	Bias%	ICP 25-75	ABW	Mark
GBM	20.22	19.3730	-17.9219	-12.62	57.9%	27.3426	5
Modified GBM	18.09	17.3793	-14.8372	-9.99	63.5%	29.3614	3
GARCH	**12.69**	**10.9539**	**+6.3423**	**+7.53**	**52.4%**	**34.5475**	**1**
Heston	20.20	19.3817	-17.7659	-12.46	55.6%	26.3950	4
Merton	20.65	19.7910	-18.4425	-13.04	57.1%	26.8815	6
GARCH–Merton	12.97	11.3942	+6.0360	+7.42	52.4%	34.4570	2
Heston–Merton	20.74	19.9039	-18.4588	-13.03	58.7%	29.6201	7

RMSE% is $100\times\sqrt{\mathrm{mean}(((p50-S_t)/S_t)^2)}$. MAE and Bias $ are in stock-price dollars. ICP 25–75 = share of S_t inside [p25, p75]. ABW = mean(p75−p25). Ranking uses RMSE% only.

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells bes
1	**Merton**	**26.52**	**21.61**	**4**
2	Heston–Merton	27.20	20.94	3
3	Heston	27.73	20.42	1
4	GBM	28.17	20.73	0
5	GARCH–Merton	31.91	19.17	2
6	GARCH	32.34	18.62	4
7	Modified GBM	34.32	25.21	2

Table S2 · Wins by company (lowest RMSE% in that company×regi

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	0	0	0	0	0
Modified GBM	1	1	0	0	2
GARCH	**0**	**1**	**2**	**1**	**4**
Heston	1	0	0	0	1
Merton	1	1	0	2	4
GARCH–Mertor	0	1	1	0	2
Heston–Mertor	1	0	1	1	3

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	43.53	21.11	30.22	17.82	28.17
Modified GBM	60.90	25.62	33.41	17.34	34.32
GARCH	54.43	21.49	41.50	11.94	32.34
Heston	43.29	20.34	29.60	17.70	27.73
Merton	**40.69**	**16.48**	**30.63**	**18.30**	**26.52**
GARCH–Merton	57.11	20.95	37.90	11.68	31.91
Heston–Merton	41.49	20.25	28.96	18.09	27.20

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Page 8 · Summary by company · mean RMSE% across four regimes

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	14.94	10.18	0
Modified GBM	**14.58**	**8.59**	**1**
GARCH	20.33	17.52	0
Heston	15.17	10.70	1
Merton	16.13	13.77	1
GARCH–Merton	19.05	14.41	0
Heston–Merton	14.90	10.63	1

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	41.45	35.52	0
Modified GBM	43.81	38.27	1
GARCH	44.86	30.89	1
Heston	41.24	34.72	0
Merton	**39.39**	**31.76**	**1**
GARCH–Merton	45.45	29.04	1
Heston–Merton	40.76	34.51	0

Table S4.3 · MSFT

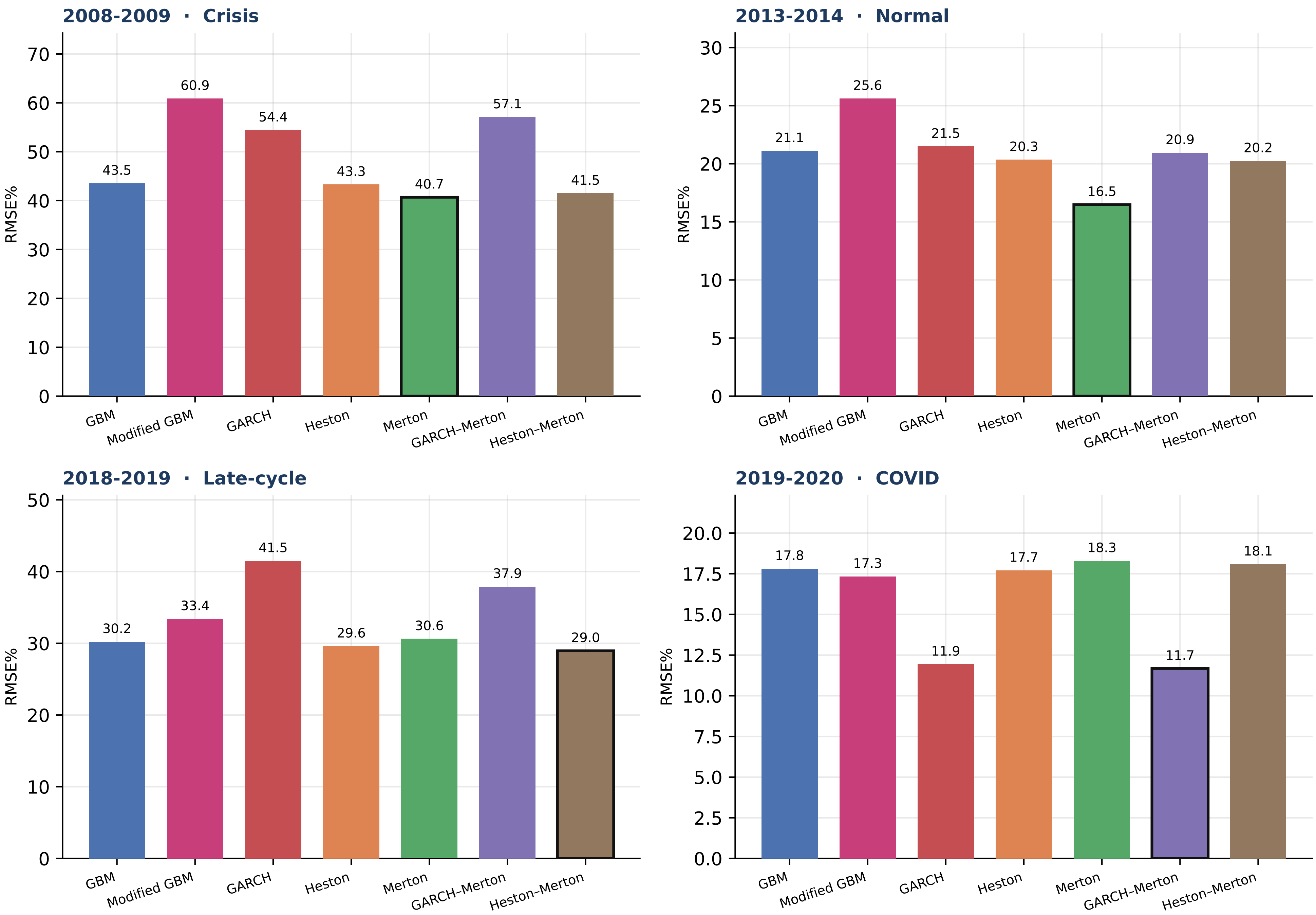
Model	Mean RMSE%	Median	# regimes best
GBM	14.27	14.33	0
Modified GBM	16.16	16.23	0
GARCH	**13.30**	**11.56**	**2**
Heston	13.88	13.97	0
Merton	15.48	16.61	0
GARCH–Merton	13.77	14.40	1
Heston–Merton	14.10	14.17	1

Table S4.4 · AMZN

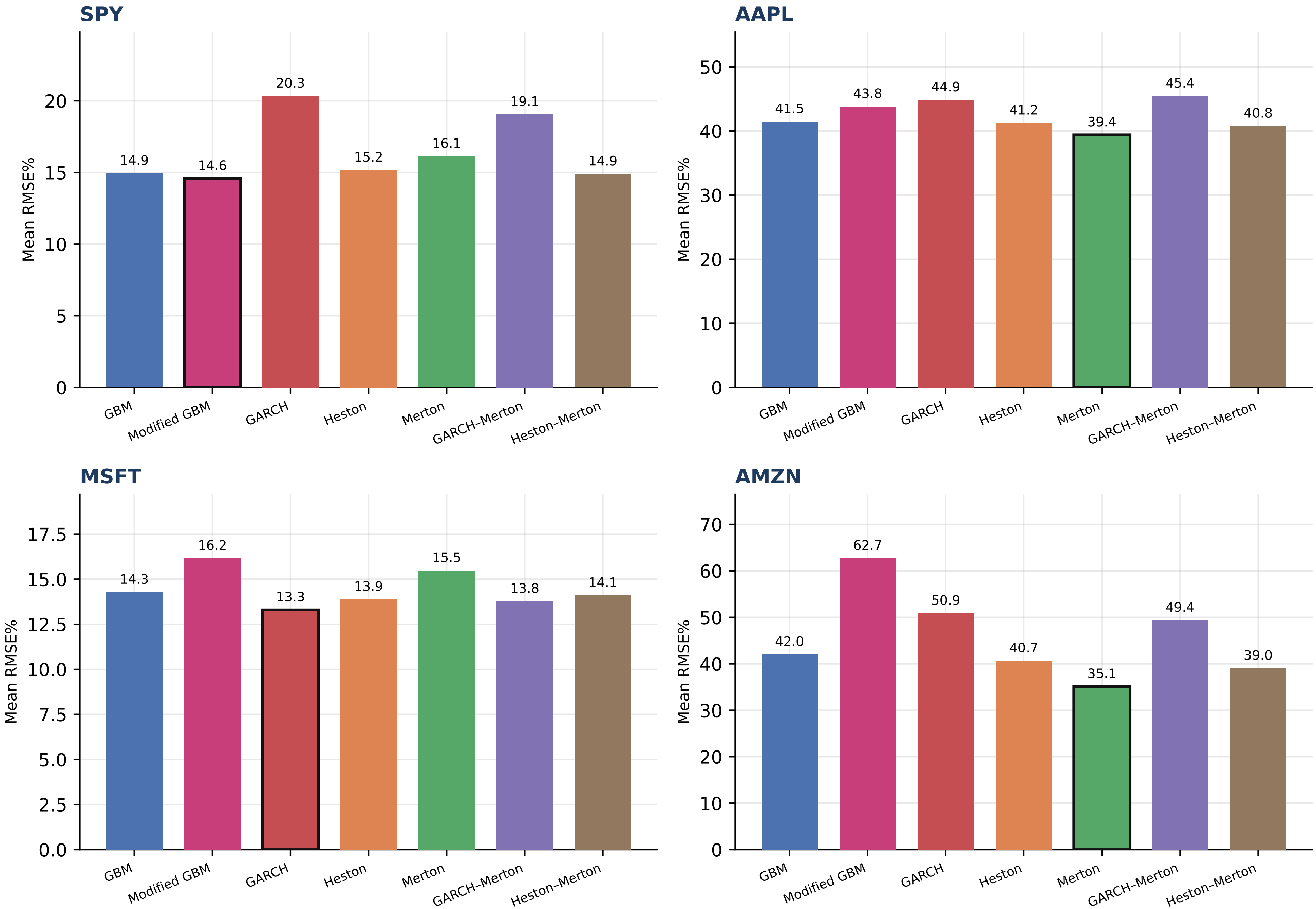
Model	Mean RMSE%	Median	# regimes best
GBM	42.00	47.98	0
Modified GBM	62.71	59.36	0
GARCH	50.86	59.24	1
Heston	40.65	46.08	0
Merton	**35.09**	**34.48**	**2**
GARCH–Merton	49.37	58.22	0
Heston–Merton	39.03	43.26	1

Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.

Page 9 · Figure 1 · RMSE% by model and regime · mean of SPY / AAPL / MSFT / AMZN

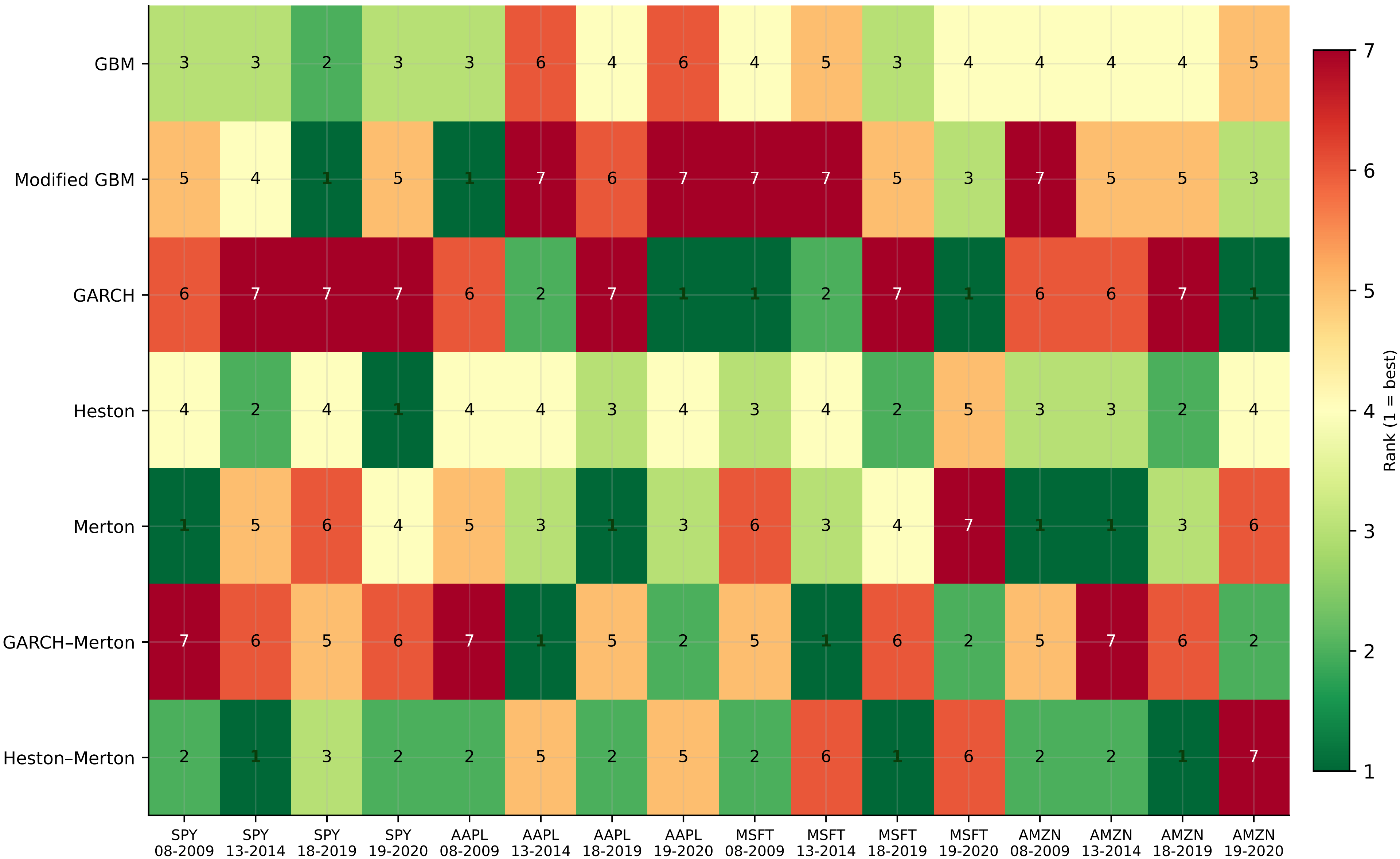


Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



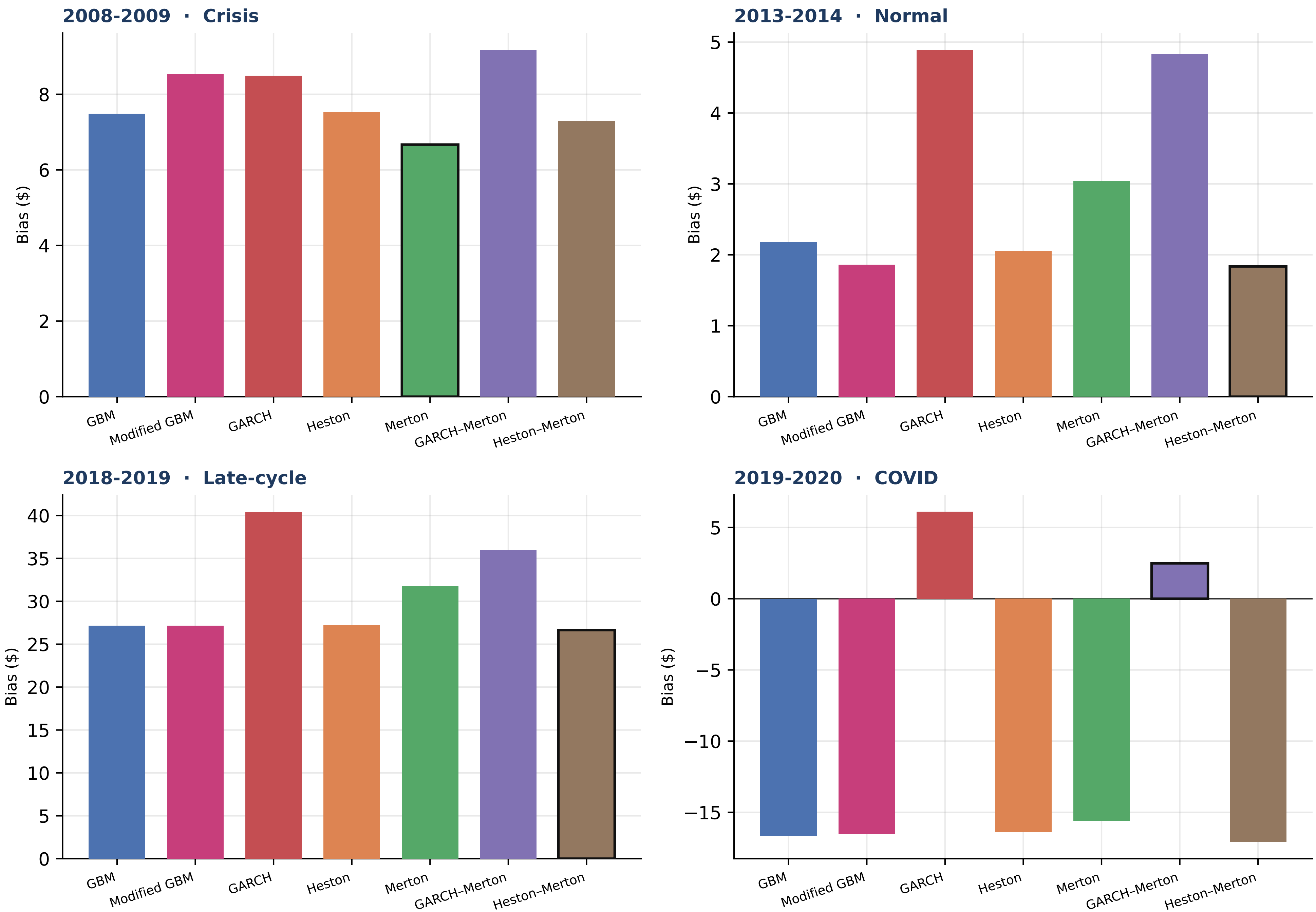
Outlined bar = lowest mean RMSE% for that company.

Page 11 · Figure 3 · Model rank by company × regime (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.

Page 12 · Figure 4 · Bias (model − market, $) · mean of four companies



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 16 company×regime cells, Merton is best (mean RMSE% = 26.52; 4 of 16 cells).
- The ranking is not stable. Best-in-cell models: GARCH, GARCH–Merton, Heston, Heston–Merton, Merton, Modified GBM.
- These conclusions are conditional on 1.5-year fixed calibration P-measure calibration, 10000 paths, daily steps, and the p50 scoring convention.

Do not mix these stock-path RMSE% numbers with the option-pricing RMSE% in the decision PDF.